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MIDTERM EXAM

Course	**:** Computational Statistics
Lecturer	**:** Muhammad Afif Hendawan, S.Kom., M.T.
Duration	**:** 150 Menit
Type	**:** Open Book

In this midterm exam, you will try to analyze ***Medical Cost Analysis*** data. Please download the data from https://p-3m.com/utsstatkom25.

Rules:

- ATTENTION! The data is formatted in ***comma separated value*** (CSV). You can convert the data to Microsoft Excel format if needed.
- You can use any analytical tool for this exam except the tool which specialized for statistical analysis, e.g. SPSS, JASP, MINITAB
- **ANY COMMUNICATION TOOL IS PROHIBITED!**

Case Study: Medical Cost Analysis

Analyze what factors influence medical insurance charges and understand data behavior using linear regression and distribution analysis.

The dataset contains 1338 records with these following columns,

1. ‘age’ : Age of beneficiary
2. ‘sex’ : Gender (Male, Female)
3. ‘bmi’ : *Body Mass Index*
4. ‘children’ : Number of children covered by insurance
5. ‘smoker’ : Smoking status
6. ‘region’ : US Region
7. ‘charges’ : Insurance cost billed to the patient

Section 1: Data Preparation and Descriptive Analysis

1. Import the dataset using your preferred tool (Google Spreadsheet, Microsoft Excel, Python, etc).
2. Explore the basic values,
   a. Mean, median, minimum, maximum, and standard deviation for ‘charges’, ‘age’, ‘bmi’. (**5 points**)
   b. Create a histogram of ‘charges’ and ‘bmi’. (**5 points**)
3. Identify the outliers in ‘charges’ by using Z-scores method.
   a. Compute the value of Z-scores.
   b. Count the values with $|Z| > 3$! (**20 points**)

Section 2: Probability Distribution

1. Assumes 'charges' is normally distributed,
    a. Calculate the probability that a person's charges **exceed** $15.000? (**10 points**)
    b. What percentage of patients have BMI below 25? (**5 points**)
2. Do actual distribution of 'charges' and 'bmi' appear normal? Justify using **boxplot** with explanation. (**5 point**)

Section 3: Regression Analysis

1. Build a simple linear regression model to predict 'charges' from 'bmi',
    a. What is the regression model based on the regression equation $\hat{y} = a + bx$? (**15 points**)
    b. How strong the relationship between 'charges' and 'bmi'? (**10 points**)
2. Build a multiple regression model to predict 'charges' using 'age', 'bmi', 'children', and 'smoker'.
    a. What is the regression model based on the regression equation
    $\hat{y} = b_0 + b_1x_1 + b_2x_2 + \cdots + b_nx_n$?
    (**10 points**)
    b. Which predictor has the strongest influence on 'charges'? Prove it by using correlation analysis!
    (**15 points**)

Hints: *If a variable is qualitative, it must first be converted into a quantitative value. This process is called encoding.*